



FORWARD MODELLING IN CROSS-HOLE SEISMIC TOMOGRAPHY USING RECIPROCITY

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Abstract—Seismic transmission tomography provides a means of direct estimation of compressional wave velocities using first arrival traveltimes from cross-hole data. Forward modelling in seismic-tomography involves computation of traveltimes at the receiver positions and estimation of ray travelpaths for a subsurface velocity model and source–receiver geometry. Conventional ray-tracing techniques face many difficulties when applied to real-life situations. These can be overcome to a great extent by using a ray-tracing technique based on the principle of reciprocity. In the present work, an algorithm is developed which makes use of the reciprocity principle and the Fermat's principle for estimation of raypaths.

In this technique seismic traveltime calculation is based on a two-dimensional dynamic programming approach, involving systematic mapping of traveltimes over a grid of constant-velocity cells. The algorithm works for even highly complicated velocity models. First arrival seismic energy can travel either as transmitted waves, diffracted waves, or head waves, and this technique simulates all of them.

The algorithm is tested for various subsurface velocity models, some of which are presented in this paper. The results are represented as ray diagrams, superimposed over the actual velocity model. The isotime lines depicting the wavefronts are also presented, which reveal the effect of velocity contrast on seismic wave propagation. © 1997 Elsevier Science Ltd. All rights reserved.

1. INTRODUCTION

Seismic tomography is a method of inversion for obtaining subsurface velocity models that adequately describe the seismic data and show the effects of rock properties on seismic wave propagation. The objective is to reconstruct the subsurface velocity structure using the principles of seismic wave propagation. Seismic tomography has been used in two models: reflection and transmission. Transmission tomography considers the seismic energy travelling through the sub-surface without reflections. This requires placing the source in a borehole and receivers at the surface or vice versa, or, the source may be placed in one borehole and the receivers in the other as shown in Fig. 1. The cross borehole model of transmission tomography has been most popular because of its ability to scan the entire region between the boreholes. A representative situation depicting ($V_p = 2200 \text{ m s}^{-1}$) intrusive body within a low velocity ($V_p = 1500 \text{ m s}^{-1}$) sedimentary formation is given in Fig. 2. Two boreholes on either side of the region contains an array of seismic sources and receivers. Given the first arrival time data, the aim is to reconstruct the velocity structure between the boreholes. To achieve this, we need to determine the

optimum ray-path between each source–receiver pair, and then solve a set of equations to obtain the velocity structures. This is generally done by simple backprojection, algebraic reconstruction techniques, or certain iterative inversion schemes like SVD, DLSQ, etc.

The procedure in cross-hole seismic tomography can be followed as:

- 1 determination of actual traveltimes at the receiver positions for the given source–receiver geometry;

- 2 ray-trace modelling of energy travelpaths;

- 3 solution of traveltime equations.

Forward modelling in seismic tomography involves the second step, where the seismic traveltime between a source and a receiver is calculated numerically for an assumed velocity model and the actual ray travelpath is traced for the given model. This is an important step in traveltime tomography and is carried out at each iteration.

The forward modelling has traditionally been done by ray-tracing. Ray-tracing is based on the concept that seismic energy follows a trajectory determined by the ray-tracing equations. Physically, these equations describe how energy continues in the same direction until it is refracted by velocity variations. Many researchers have developed ray-tracing techniques [1–3]. There are two types of ray-tracing techniques: the *shooting* ray method and the *bending* ray method. The shooting ray method involves specification of a single

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raypath by a shooting angle and calculation of traveltime along this raypath and the bending ray method involves bending of raypath by velocity perturbations until it satisfies a minimum traveltime criterion. Thurber and Ellsworth [4] discussed these methods in more detail.

The ray-tracing method faces three major difficulties. First, for strongly varying velocity fields, there may be many paths connecting the two points of interest and there is a chance of missing out the minimum time path. Second, computer cost escalates rapidly for greater numbers of cells, making the method impractical. Third, the linear relationship assumed between the shooting angle and the end point of the raypath may not be valid for complicated velocity structures.

To overcome these difficulties, many researchers have come up with alternative approaches to seismic traveltime calculation. These methods are based on finite difference calculations [5, 6], Huygen's principle [7], graph theory [8], etc.

In the present work, the problem is divided into two different steps: first, calculation of traveltimes at all the grid points and second, estimation of the raypath using reciprocity principle on the traveltime values already computed [9]. This is a reverse approach as compared to the conventional ray-tracing algorithm, where the raypath is determined first and the traveltime is calculated by integrating the traveltime values over the raypath. The advantage here is that we don't have to do a hit and trial method as undertaken in conventional ray-tracing, and also, once the traveltimes are computed the minimum time path can be determined uniquely. A traveltime calculation scheme based on two-dimensional dynamic programming approach [10] is used for computing the first arrival times over the grid.

The algorithm is tested numerically for various velocity structures on a large domain of 512×512 grid points. Some of the results are presented in this paper. It is implemented in HP-FORTRAN77 in the HP-UX 7.0 environment on HP-9000/835 workstation and is also tested on DEC AXP-3000 system in the OSF/1 V3.0 environment.

2. METHODOLOGY AND ALGORITHM

The ray-tracing problem can be divided into two different mathematical problems: one is to evaluate the traveltimes for a given subsurface velocity model, and the other is to estimate the raypath from these traveltime data. An advantage in this type of approach is that any suitable method can be used for calculation of traveltimes depending on the velocity model at hand. In the present work we have used a dynamic programming approach to traveltime calculations over a grid, which divides the subsurface into an array of constant velocity cells. The details of this approach are discussed here. Once we have the traveltime data, the raypath is estimated by applying

the two basic principles of seismic wave propagation, these are:

- *Acoustic reciprocity principle*—the path of wave propagation from a source S to a receiver R is the same as that from a source R to a receiver S, as shown in Fig. 3.
- *Fermat's principle*—the traveltime of seismic raypath between two points is the minimum compared to all the neighbouring paths.

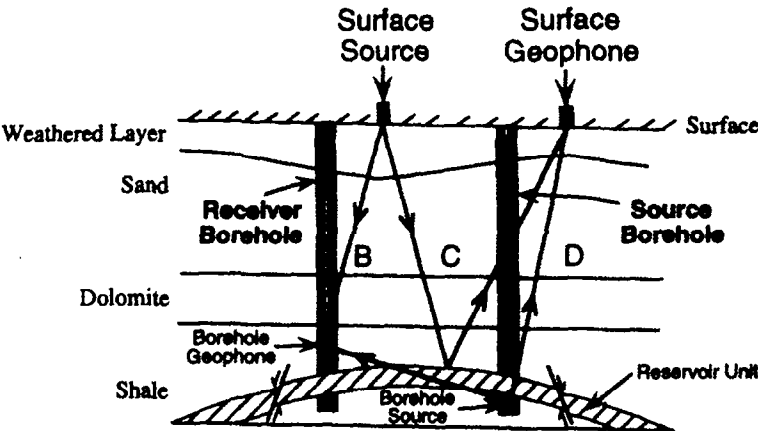
These two principles are combined to develop an algorithm for seismic traveltime calculation and raypath estimation. The algorithm has the following steps:

- (1) The minimum traveltimes from a source point S to all grid points in the cell model are computed.
- (2) The source and the receiver positions are interchanged and the minimum traveltimes at all the grid points are calculated in the reverse direction.
- (3) The traveltime data obtained from the above two steps are added to obtain a two-way traveltime map over the entire mesh.
- (4) Once the two-way traveltimes are obtained for all the grid points, Fermat's principle is used to estimate the raypath from the global minimum traveltimes.

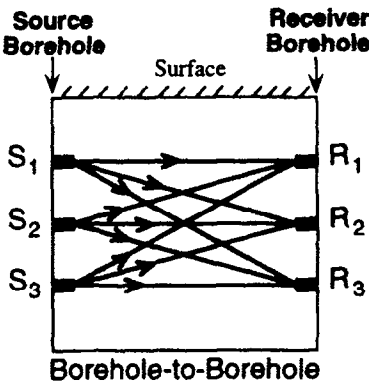
In Fig. 3, the central curve is the desired raypath. The total traveltime on this path is the minimum value $\tau = \tau_1 + \tau_2$. The traveltimes along neighbouring paths are always larger than the traveltime τ of the desired path by δt ; i.e. $\tau' = \tau + \delta t$.

The procedure applied for the calculation of traveltimes makes use of a dynamic programming approach, where the traveltimes are mapped in a markovian process. Given the geometry of the mesh, for calculating the first arrival time at any point (R), the arrival times at only the three neighbouring points are required. Two of these points (H,E) are located on the previous column and one (G) is located on the same column. The situation is illustrated for cell B in Fig. 4. The mapping begins by calculating the arrival times over the left edge of the grid. The arrival time at R is the minimum of traveltimes H to R, E to R, F_{row} to R and G to R. The values obtained at each point on a column are then stored in a floating array, which is then used for calculating the values on the next column. After the values on the next column are computed, the floating array is updated by substituting the old values by the new ones. The mapping thus shifts by one column and the computation over the next column starts. The process continues until we reach the right edge of the grid. At this stage we have the first arrival times at all the grid points starting from the source. The process is then continued in the reverse direction, i.e. by interchanging the source and the receiver positions. The same floating array is again updated so that the time value at any point now becomes the time taken for the ray to travel from the

(a) Ray Paths in Tomography



(b) Transmission Tomography Geometry: Borehole-to-Borehole



(c) Transmission Tomography Geometry: Surface-to-Borehole

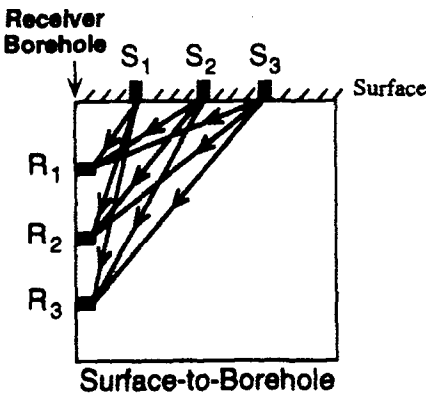


Fig. 1. Seismic travelttime tomography models.

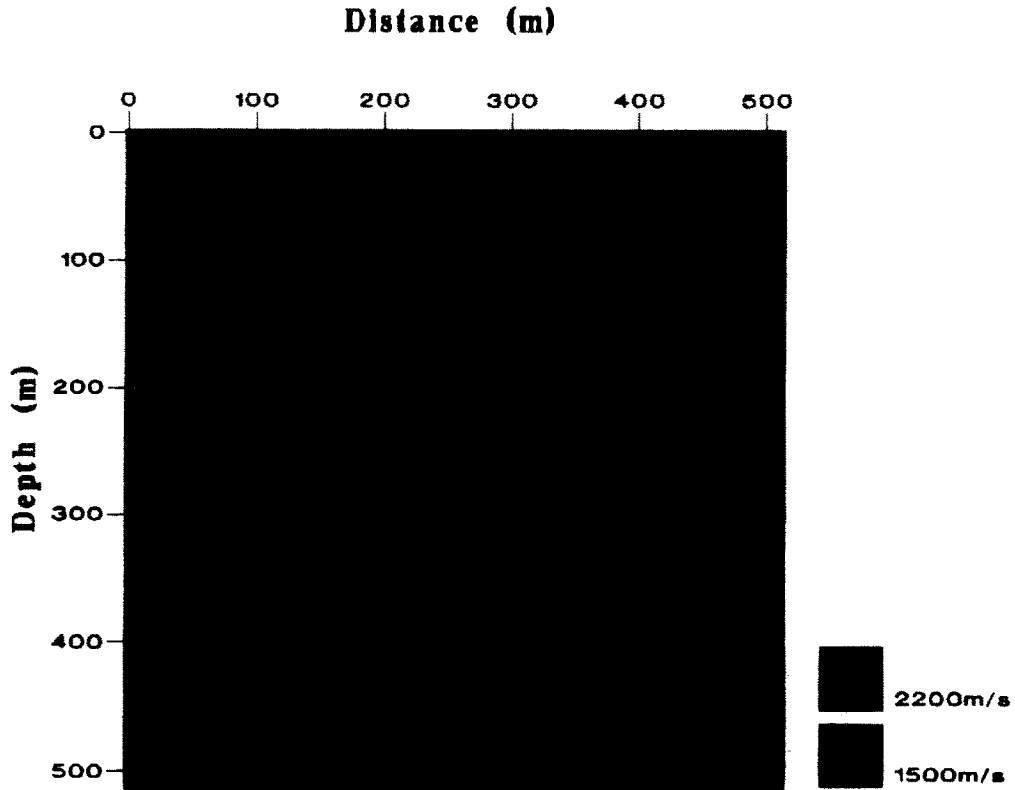


Fig. 2. A typical cross-hole tomographic representation of a high velocity intrusive body.

source to the receiver, and then back from the receiver to the point under consideration. Thus the total time at any point becomes the two-way traveltime for the ray passing through that point.

In the above procedure the most important step is the computation of the first arrival time at a grid point, given the first arrival times at the three

neighbouring points. To achieve this we apply two interpolation schemes. The first one is the *linear interpolation scheme* (based on a straight raypath assumption) and the second is the *nonlinear interpolation scheme* (for curved raypaths). Travel-time calculation schemes using these two interpolation methods are discussed below.

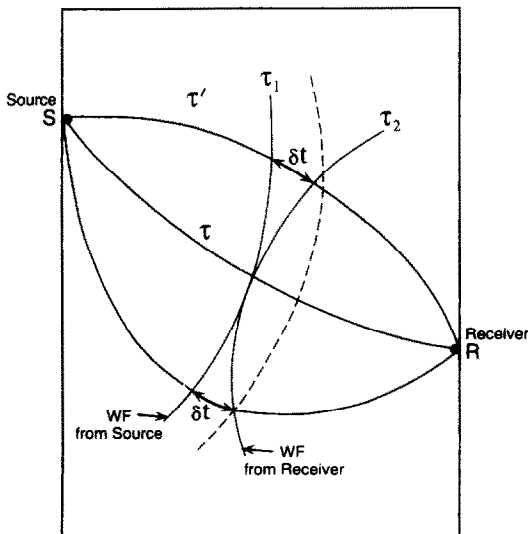


Fig. 3. Tracing of shortest path by Fermat's principle.

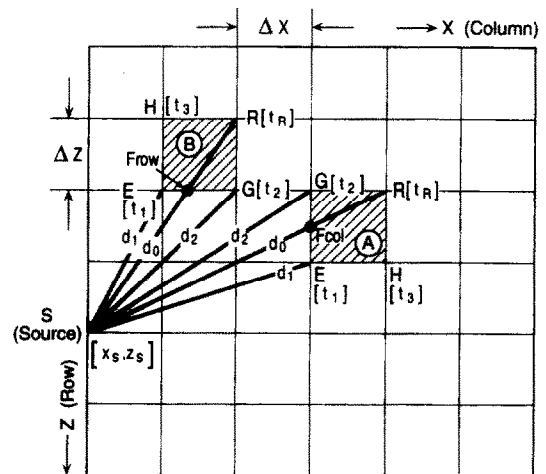


Fig. 4. Grid layout for traveltime calculations using dynamic programming (notice that the grid points G and H interchange positions for the computation of row and column intersection points F_{row} and F_{col} as depicted in the cells B and A, respectively).

2.1. Linear interpolation method

The linear interpolation method is based on a local plane assumption, i.e. the traveltimes varies linearly with distance in any direction. For calculating the first arrival time at any grid point R over the mesh, we need the times t_1 , t_2 and t_3 at the three neighbouring points as shown in Fig. 4. We can have two different situations: (a) the straight line joining R and the source S intersects the nearest column, and (b) it intersects the nearest row. Let t_0 be the arrival time at the point of intersection, row or column. If d_1 , d_2 and d_0 are the corresponding linear distances, of the points as shown in the figure for either cell B (depicting intersection with row) or cell A (depicting intersection with column), then by using a plane wave assumption, the time t_0 can be approximated as:

$$t_0 = t_1 + \frac{d_0 - d_1}{d_2 - d_1} (t_2 - t_1). \quad (1)$$

Given the geometry of the grid and the position of the source, the values d_0 , d_1 and d_2 are calculated and hence the time t_0 is computed using eqn (1). The next step is to compute the times at R for all the three rays travelling through three different points under consideration.

If s be the slowness of the cell (say B), then the time taken for the ray through the first grid point is given by,

$$tt_2 = t_2 + s\Delta z \quad (2)$$

for the path G to R and that for the path H to R it is given as,

$$tt_3 = t_3 + s\Delta x \quad (3)$$

and the time taken through the intersection point (row or column) is,

$$tt_0 = t_0 + s(d - d_0) \quad (4)$$

where d is the linear distance between the receiver R and the source S. Therefore, by Fermat's principle the first arrival time at R should be the minimum of the traveltimes for all the raypaths. Thus the first arrival time at R is simply the minimum of tt_2 , tt_3 and tt_0 .

The process is repeated for all the grid points over the entire mesh in a systematic manner to obtain the traveltimes map over the grid.

2.2. Nonlinear interpolation

The linear interpolation method for traveltimes calculation as described above works well for velocity structures of less or moderate complexity and low contrast but fails for very complicated structures. For such cases, a non-linear interpolation method is used. Like the linear method, the traveltimes map is

prepared over the entire grid space but in this case the traveltimes at each grid point is calculated by considering four time values, two from the neighbouring grid points while the other two time values are calculated by considering the intersection of the raypath with the nearest row and the column. The points of intersection are determined by the root bisection method. The basic theory involved in this traveltimes computation method for the rays travelling through the grid space are discussed below.

(1) *Intersection with rows.* With reference to the geometry of cell B in Fig. 5a, d_1 , d_0 , d_2 , the distances from the source S to the corner points E, F, G are given as,

$$d_1 = \sqrt{(x_s - x_1)^2 + (z_s - z_1)^2} \quad (5)$$

$$d_2 = \sqrt{(x_s - x_2)^2 + (z_s - z_1)^2} \quad (6)$$

$$d_0 = \sqrt{(x_s - x_0)^2 + (z_s - z_1)^2}. \quad (7)$$

If $\bar{s}_b$ be the average slowness between the source S and the cell B, then

$$t_1^2 = \bar{s}_b^2 d_1^2; \quad t_2^2 = \bar{s}_b^2 d_2^2$$

$$(t_2^2 - t_1^2) = \bar{s}_b^2 (d_2^2 - d_1^2)$$

$$\bar{s}_b^2 = \frac{(t_2^2 - t_1^2)}{(x_s - x_2)^2 - (x_s - x_1)^2} = W(\text{say})$$

where, t_1 , t_2 are the times of raypaths SE and SG, respectively.

Knowing t_1 , t_2 , x_2 , x_1 , x_s , W can be calculated. Again,

$$t_2^2 = t_1^2 + W((x_s - x_2)^2 + (x_s - x_1)^2). \quad (8)$$

Let the traveltimes at R along FR be t and the time taken by the way to reach F from the source be t_0 , then

$$t = t_0 + s_b \sqrt{((\Delta z)^2 + (x_2 - x_0)^2)} \quad (9)$$

where s_b is the slowness in the cell B. Therefore,

$$t_0^2 = t^2 + W((x_0 - x_s)^2 + (x_1 - x_s)^2). \quad (10)$$

Differentiating eqn (9) with respect to x_0 , we get,

$$\frac{dt_0}{dx_0} = \frac{dt}{dx_0} + \frac{s_b}{\Delta z^2 + (x_2 - x_0)^2} (x_0 - x_2). \quad (11)$$

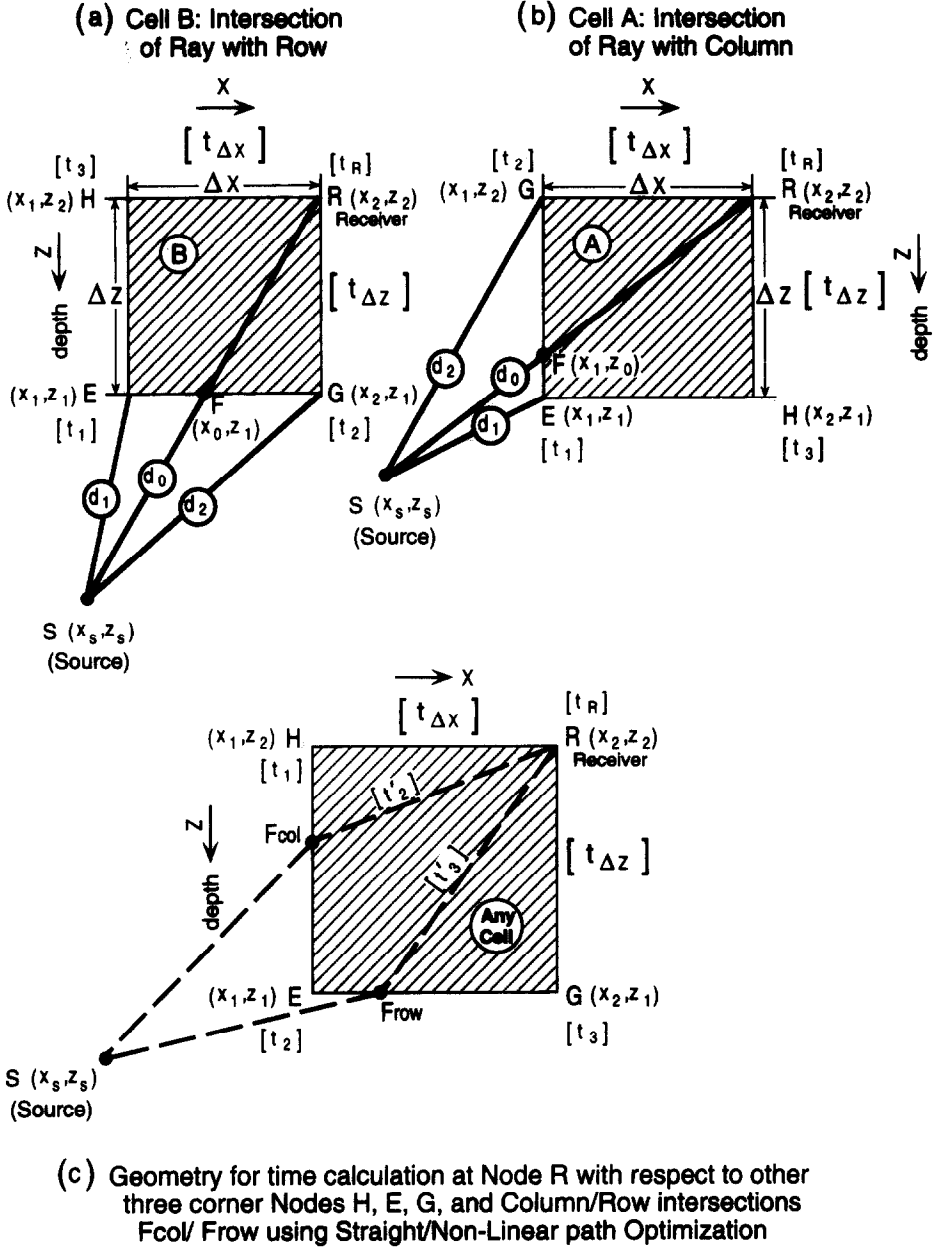


Fig. 5. Implementation of non-linear interpolation scheme for calculation of first arrival times.

Differentiating eqn (10) with respect to x_0 , we get,

$$2t_0 \frac{dt_0}{dx_0} = 2W(x_0 - x_s) \quad (12)$$

or,

$$\frac{dt_0}{dx_0} = \frac{W}{t_0} (x_0 - x_s). \quad (13)$$

Substituting the value of dt_0/dx_0 in eqn (11) we get,

$$\frac{dt}{dx_0} = \frac{W}{t_0} (x_0 - x_s) + \frac{s_b}{\Delta z^2 + (x_2 - x_0)^2} (x_0 - x_2). \quad (14)$$

For t to be minimum, dt/dx_0 should be equal to zero. Therefore,

$$\frac{s_b^2 (x_2 - x_0)^2}{\Delta z^2 + (x_0 - x_2)^2} = \frac{W^2 (x_0 - x_s)^2}{t_0^2}. \quad (15)$$

Thus we can now define a function FUNC_{row} as,

$$\text{FUNC}_{\text{row}} = \frac{s_b^2 (x_2 - x_0)^2}{\Delta z^2 + (x_0 - x_2)^2} - \frac{W^2 (x_0 - x_s)^2}{t_0^2}. \quad (16)$$

(2) *Intersection with column.* With reference to cell A in Fig. 5b, d_1 , d_0 , d_2 , the distances from the source S to the corner points E, F, G can be written as,

$$d_1 = \sqrt{(x_s - x_1)^2 + (z_s - z_1)^2} \quad (17)$$

$$d_2 = \sqrt{(x_s - x_1)^2 + (z_s - z_2)^2} \quad (18)$$

$$d_0 = \sqrt{(x_s - x_1)^2 + (z_s - z_0)^2}. \quad (19)$$

If $\bar{s}_a$ be the average slowness between the source and the cell A, then

$$t_1^2 = \bar{s}_a^2 d_1^2; \quad t_2^2 = \bar{s}_a^2 d_2^2$$

$$(t_2^2 - t_1^2) = \bar{s}_a^2 (d_2^2 - d_1^2)$$

$$\bar{s}_a^2 = \frac{(t_2^2 - t_1^2)}{(z_s - z_2)^2 - (z_s - z_1)^2} = W(\text{say})$$

where t_1 , t_2 are the times of the raypaths SE and SG, respectively.

Knowing t_1 , t_2 , z_1 , z_2 , z_s , W can be calculated. Again,

$$t_0^2 = t_1^2 + W((z_s - z_2)^2 + (z_s - z_1)^2). \quad (20)$$

Let the travel time at R along FR be t and the time taken by the ray to reach F from the source be t_0 . So,

$$t = t_0 + s_a \sqrt{(\Delta x)^2 + (z_s - z_0)^2} \quad (21)$$

where s_a is the slowness in the cell A. Therefore,

$$t_0^2 = t_1^2 + W((z_0 - z_s)^2 + (z_s - z_1)^2). \quad (22)$$

Differentiating eqn (21) with respect to z_0 , we get

$$\frac{dt}{dz_0} = \frac{dt_0}{dz_0} + \frac{s_a}{\Delta x^2 + (z_s - z_0)^2} (z_0 - z_s). \quad (23)$$

From eqn (22), differentiating with respect to z_0 , we get

$$2t_0 \frac{dt_0}{dz_0} = 2W(z_0 - z_s) \quad (24)$$

or,

$$\frac{dt_0}{dz_0} = \frac{W}{t_0} (z_0 - z_s). \quad (25)$$

Substituting the value of dt_0/dz_0 in eqn (23), we get

$$\frac{dt}{dz_0} = \frac{W}{t_0} (z_0 - z_s) + \frac{s_a}{\Delta x^2 + (z_s - z_0)^2} (z_0 - z_s). \quad (26)$$

For t to be minimum, dt/dz_0 should be equal to zero. Therefore,

$$\frac{s_a^2 (z_2 - z_0)^2}{\Delta x^2 + (z_0 - z_2)^2} = \frac{W^2 (z_0 - z_s)^2}{t_0^2}. \quad (27)$$

Thus, we can also define a function FUNC_{col} as,

$$\text{FUNC}_{\text{col}} = \frac{s_a^2 (z_2 - z_0)^2}{\Delta x^2 + (z_0 - z_2)^2} - \frac{W^2 (z_0 - z_s)^2}{t_0^2}. \quad (28)$$

Combining the above two cases, the generalized approach for any cell as shown in Fig. 5c can be given as,

$$\text{FUNC}_{\text{row}} = \frac{s^2 (x_2 - x_0)^2}{\Delta x^2 + (x_0 - x_2)^2} - \frac{W^2 (x_0 - x_s)^2}{t_0^2} \quad (29)$$

and

$$\text{FUNC}_{\text{col}} = \frac{s^2 (z_2 - z_0)^2}{\Delta x^2 + (z_0 - z_2)^2} - \frac{W^2 (z_0 - z_s)^2}{t_0^2}, \quad (30)$$

where s is the slowness of the cell under consideration.

By equating FUNC_{row} and FUNC_{col} to zero, we obtain the co-ordinates of the intersection points F_{row} and F_{col} , respectively. We can then, compute the values of $t_{F_{\text{row}}}$ and $t_{F_{\text{col}}}$. These are the time values at R for rays travelling through F_{row} and F_{col} , respectively. Once we have the time values $t_1 + t_{\Delta x}$, $t_2 + t_{\Delta x}$, $t_{F_{\text{col}}}$ and $t_{F_{\text{row}}}$ for these four different raypaths, the first arrival time at R is obtained by taking the minimum of these four time values.

3. NUMERICAL MODEL STUDIES AND DISCUSSION

The following synthetic model studies prove analytically the accuracy of the forward modelling scheme using reciprocity principle as described in the earlier sections. All the velocity models studied consist of high contrast velocities separated by distinct boundaries (nonsmooth) and represent the most difficult situations for any forward modelling algorithm. The testing is done on a large domain of 512×512 cells, each cell representing a square area of size 1×1 m. Thus the region of interest extends to 512 m in both horizontal and vertical directions. Each model is tested for both linear and nonlinear optimization schemes. The resulting ray diagrams are presented by superimposing them over the corresponding velocity model. The wave propagation for each case is depicted in the snapshot obtained by isotime mapping with different grey levels. This is done only for one source and several receivers using the nonlinear optimization scheme.

3.1. Three-layered horizontal earth model

Figure 6a represents the ray-tracing for a three-layered earth model with the compressional

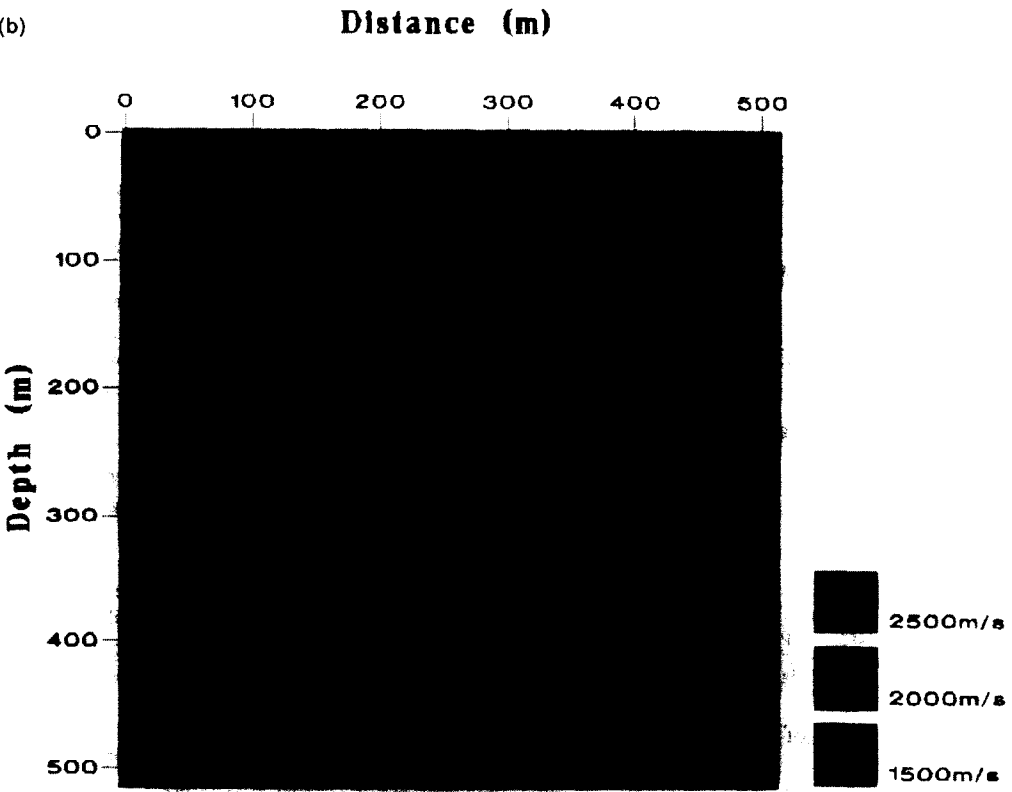
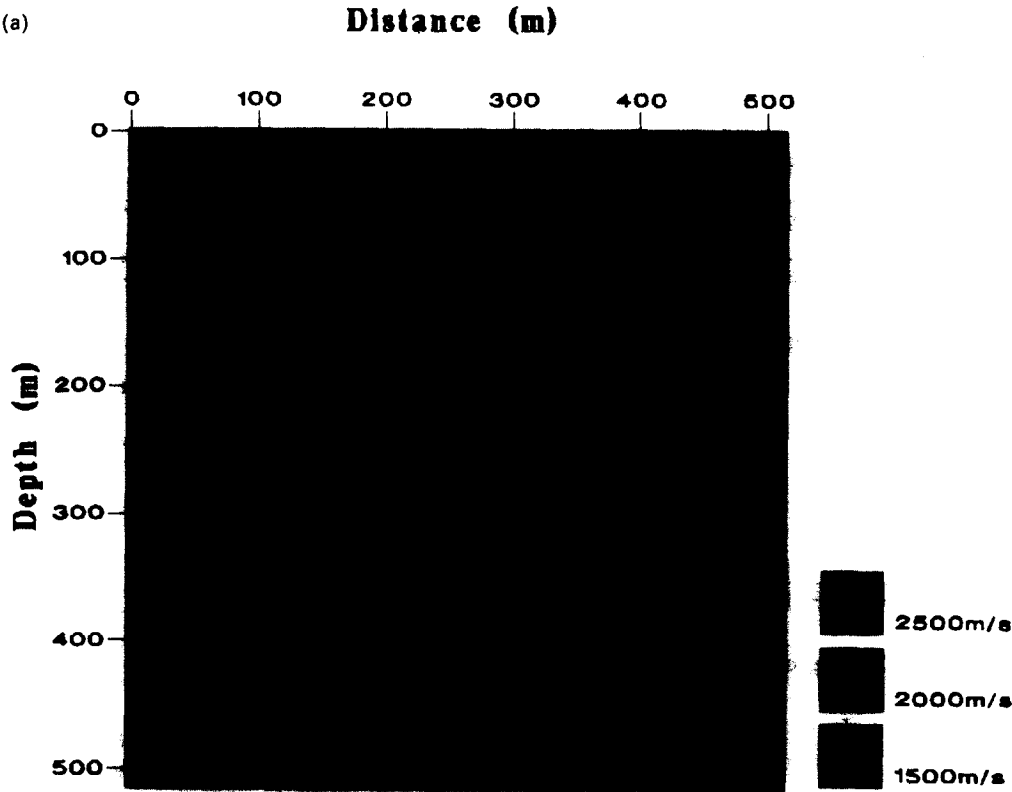


Fig. 6 (a and b)

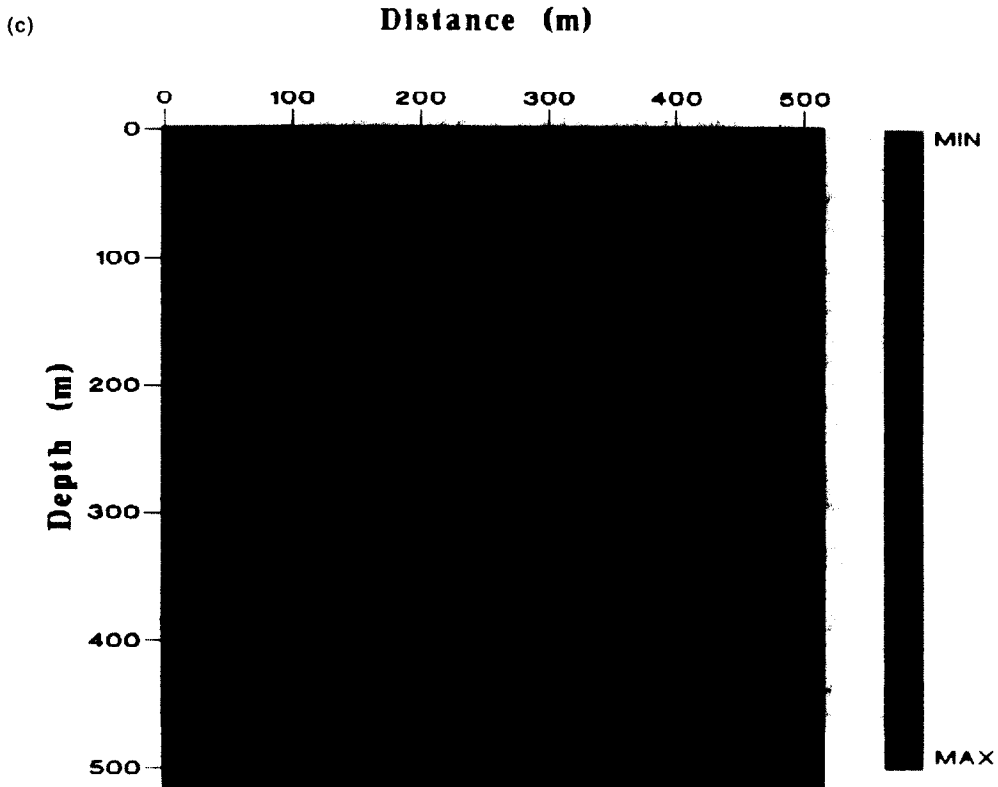


Fig. 6. (a) Physical configuration and superimposed ray diagram for a three layered earth model using linear interpolation scheme. (b) Physical configuration and superimposed ray diagram for a three layered earth model using nonlinear interpolation scheme. (c) Wave propagation snapshot of the model depicted in (b) for one source and three receivers.

wave velocities 1500 m s^{-1} in the top layer, 2000 m s^{-1} in the middle layer and 2500 m s^{-1} in the bottom layer. The three sources and receivers are placed at depth levels of 50, 250 and 450 m. The simulation is done by using the linear interpolation method. Because of this approximation, the resulting ray diagram could not be well resolved. However, the bending of the rays are observed at the interfaces separating two acoustically different media.

The ray-tracing for the same model is done by using nonlinear interpolation of the raypath and the resulting ray diagram is presented in Fig. 6b. The raypaths are now well resolved and are in accordance with Snell's law of refraction.

A snapshot depicting the wave propagation from the source at 250 m depth level below the surface, along with the superimposed raypath for the above receiver positions is given in Fig. 6c. The direct and the head wavefronts in the bottom high velocity layer are on a leading phase compared to the ones propagating in the top and the middle layers.

3.2. Two-dimensional intrusive high velocity square body

Figure 7a depicts an intrusive square body with P-wave velocity 2000 m s^{-1} in a sedimentary formation with a velocity of 1500 m s^{-1} . The ray diagram as obtained for four sources and four receivers, by using the linear interpolation scheme is presented in the same figure. The nonlinear interpolation results are presented in Fig. 7b. While the linear raypath assumption has added anomalous ray geometry, the nonlinear optimization has resolved the raypath in a much better fashion. As is expected, the rays are clustered in the high velocity intrusive body.

A snapshot for the shot point at 300 m depth and four receiver positions marked on Fig. 7c represents the wave propagation through this intrusive model. For the identification of the direct and head wavefronts, the raypaths are also superimposed on the wavefronts.

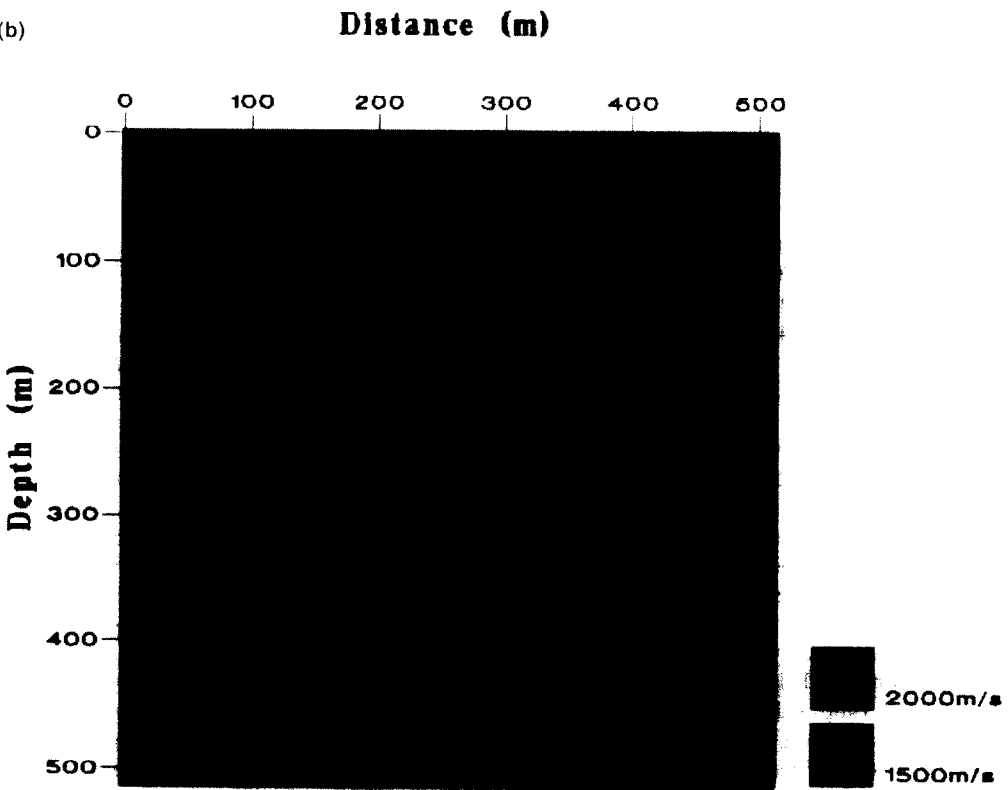
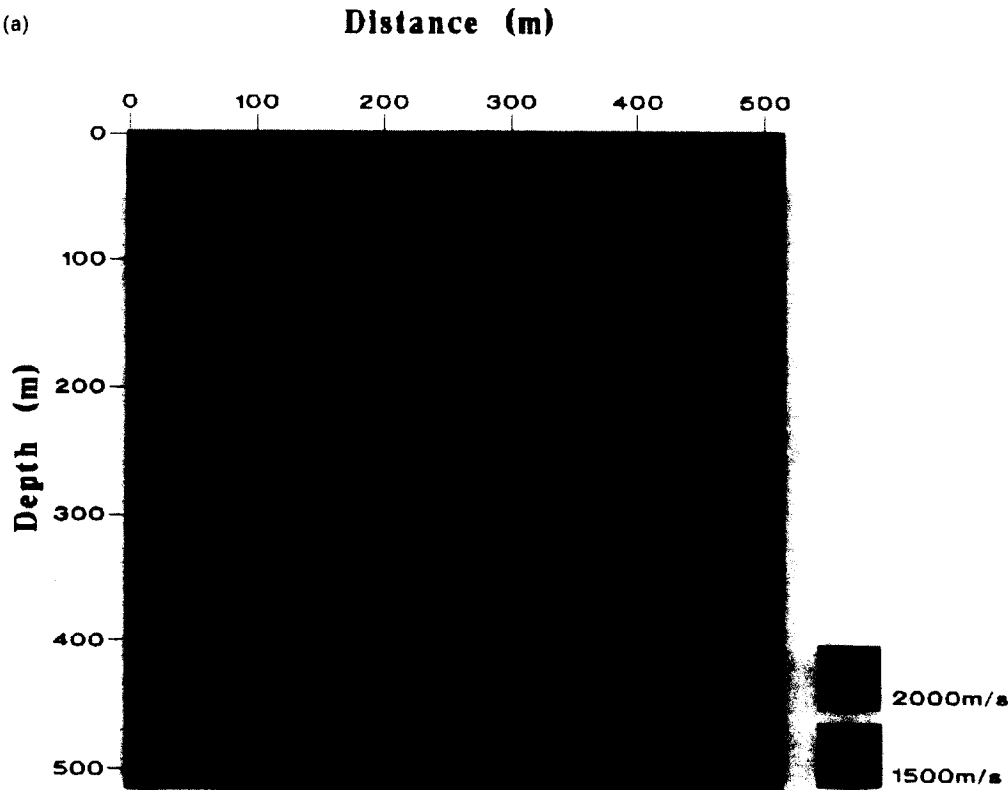


Fig. 7 (a and b)

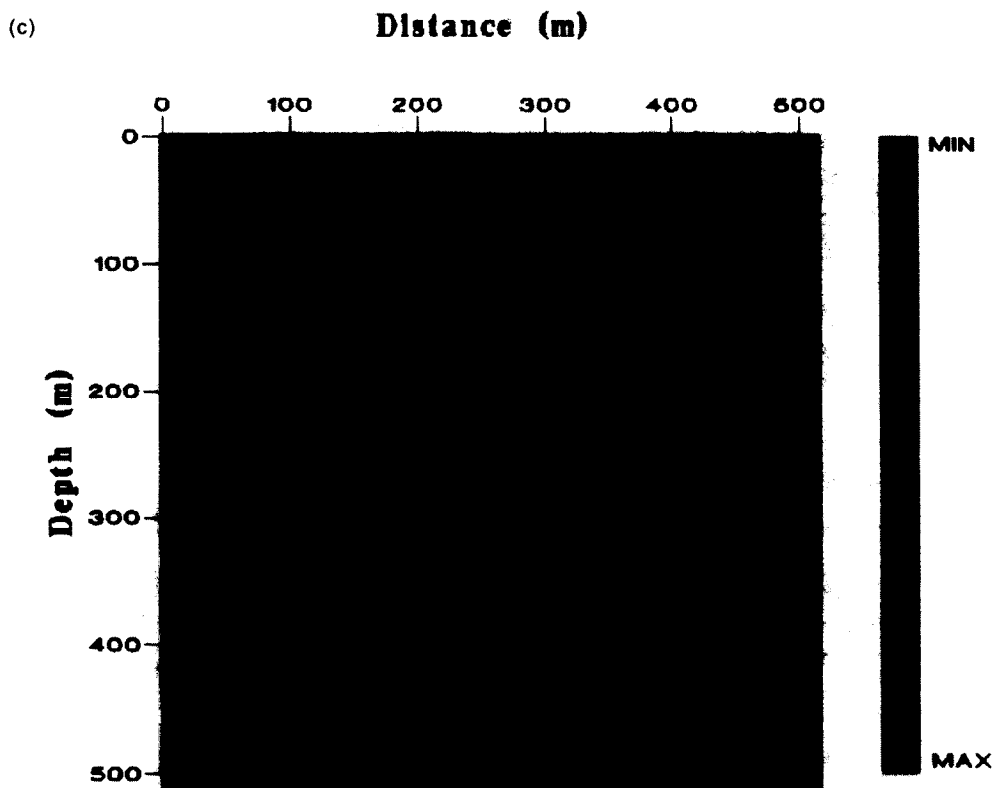


Fig. 7. (a) Physical configuration and superimposed ray diagram for a high velocity intrusive square model using linear interpolation scheme. (b) Physical configuration and superimposed ray diagram for a high velocity intrusive square model using nonlinear interpolation scheme. (c) Wave propagation snapshot of the model depicted in (b) for one source and four receivers.

3.3. A high velocity intrusive body in a three-layered earth model

Figure 8a represents the physical model along with the respective P-wave velocity in each layer. The top and the bottom layers have a P-wave velocity of 4000 m s^{-1} . The compressional wave velocity in the vertical prismatic body is 4500 m s^{-1} and that in the middle layer is 3500 m s^{-1} . This figure also depicts the ray diagram for four sources and four receivers with linear interpolation of the raypath, thereby showing anomalous ray geometry. The same model, when simulated using nonlinear optimization scheme, as shown in Fig. 8b, shows the actual raypath. Due to the cumulative effect of the otherwise infinitesimal floating error in the optimization process only at the greater angle of incidence or emergence of a ray in such a huge computational space of 512×512 grid, we at times notice minor kinks in the otherwise straight raypath in the higher velocity medium. The snapshot for this complicated subsurface model with shot point at 250 m depth is presented in Fig. 8c. The wavefronts are easily identified and the superimposed rays for four receiver positions demark the change over of the wavefronts from the direct to the head wave phase.

3.4. A void in a coal seam

Figure 9a represents the physical model of a void with irregular boundary having the P-wave air velocity of 330 m s^{-1} in a coal seam with compressional wave velocity of 2000 m s^{-1} . The void is actually digitized and incorporated in the model space instead of numerically simulating a regular structure to represent a real life situation wherein by cross-hole seismic tomography such voids can be accurately detected. Ray-tracing for four sources and four receivers is performed using the reciprocity algorithm with both linear and nonlinear interpolation schemes and the resulting ray diagrams are presented in Fig. 9a and b, respectively, superimposed on the model. As expected from the previous case studies, the actual raypath is better simulated by nonlinear optimization algorithm. Here too, the floating error in locating the intersection point by optimization process at larger angle of incidence or emergence of a ray in such a huge computation space, has attributed to minor kinks in few raypaths in the higher velocity medium. Another important aspect to be kept in mind is the point to point variation of the radius of curvature of the void boundary. It is to be noted in both the figures that not a single ray has

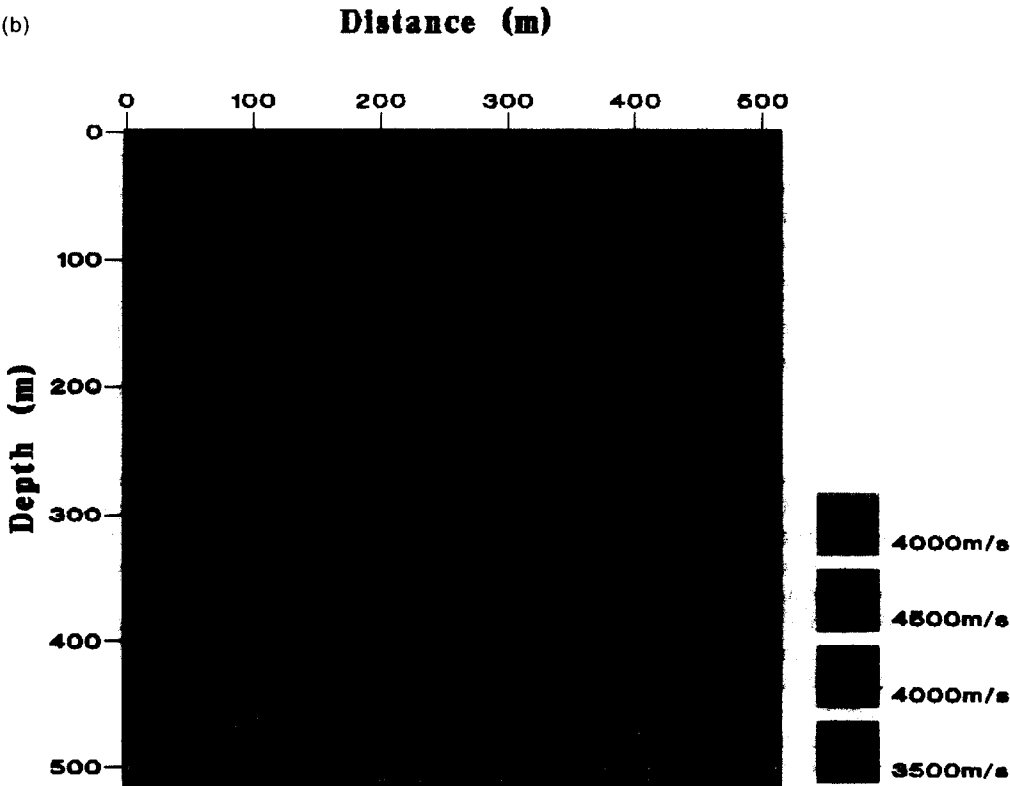
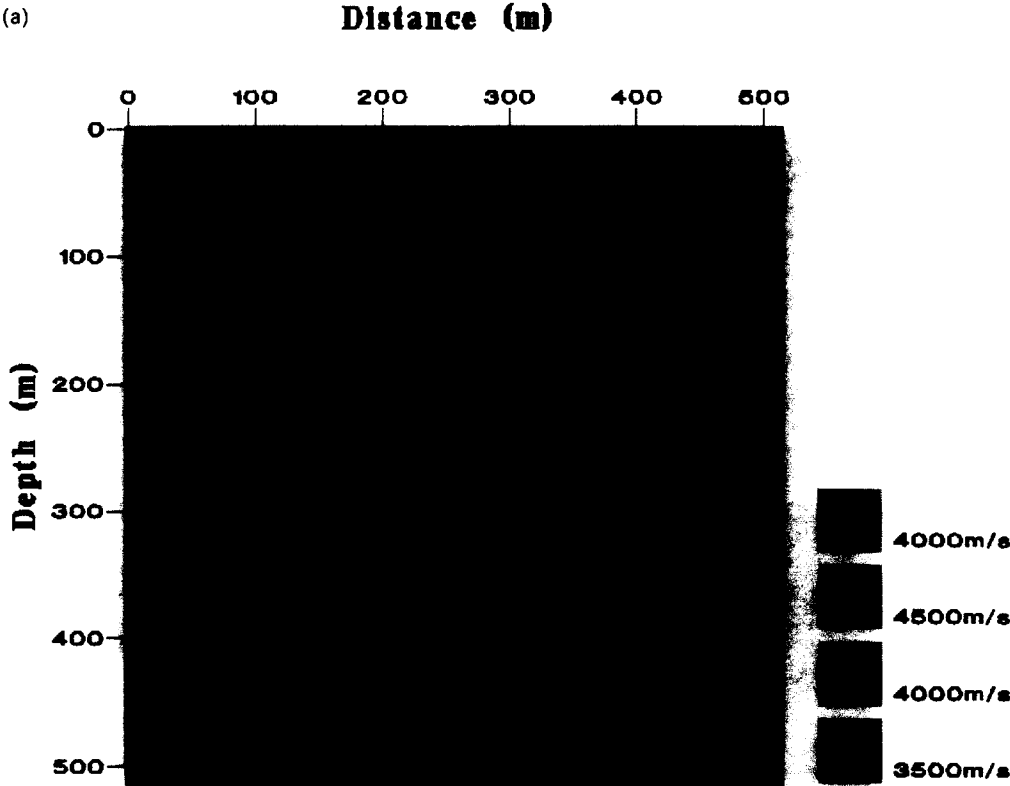


Fig. 8 (a and b)

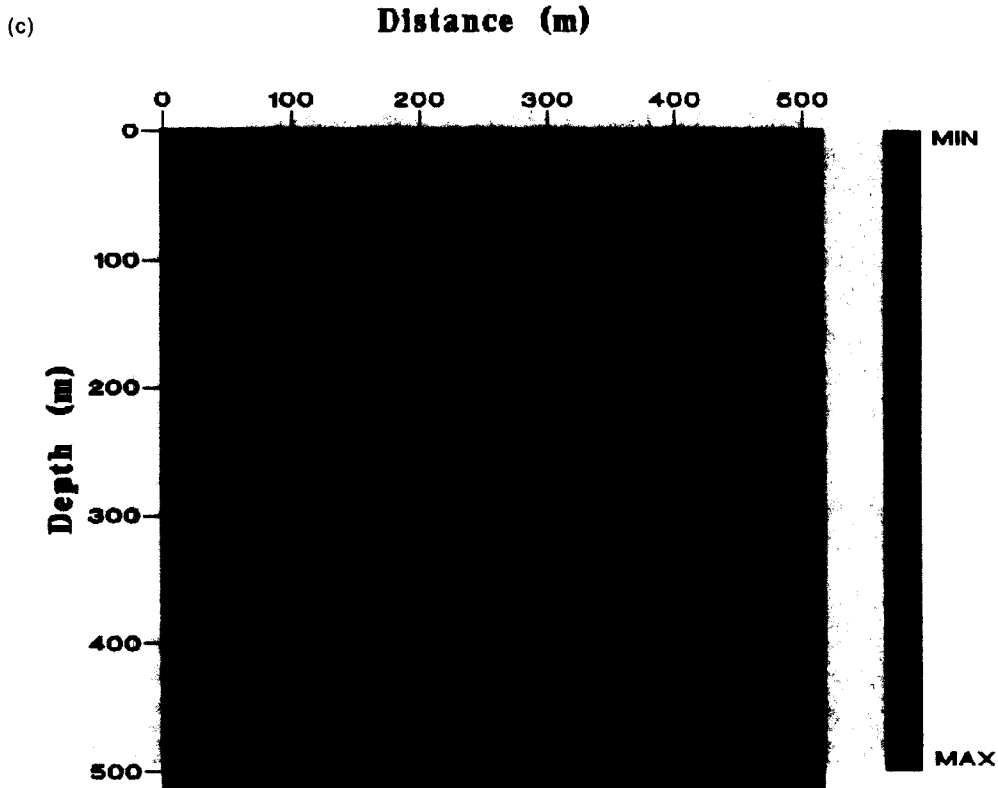


Fig. 8. (a) Physical configuration and superimposed ray diagram for a high velocity intrusive body in a three-layered earth model using linear interpolation scheme. (b) Physical configuration and superimposed ray diagram for a high velocity intrusive body in a three-layered earth model using nonlinear interpolation scheme. (c) Wave propagation snapshot of the higher velocity intrusive model of (b) for one source and four receivers.

transmitted through the void as it is optically the minimum velocity medium. The snapshot for the wave propagation through this model is presented in Fig. 9c for a single shot and four receivers. It is interestingly noticed that the diffracted wave-front surrounding the void is encompassed by the raypaths.

A closer look at these ray models along with the snapshots of the wave propagation helps us to establish the validity and the accuracy of the above forward modelling algorithm based on reciprocity principle. The results of nonlinear interpolation scheme are specially encouraging. It is interesting to note that the algorithm works for almost any kind of velocity model without much expense of computational time. For a grid size of 512×512 cells it took a CPU time of 38.8 s on an HP-900/835 workstation and 21.6 s on a DEC AXP-3000 workstation for the model given in Fig. 8b with nonlinear interpolation.

4. CONCLUSION

The ray-tracing algorithm based on reciprocity and Fermat's principle using the dynamic programming

traveltime computation approach is presented in this paper. The key of this forward modelling technique is that the minimum of the total traveltime map represents the raypaths of first arrival signals. This technique could not only overcome the difficulties that exist in the conventional forward ray-tracing methods, but also could accurately simulate the ray diagram for arbitrary, discrete and high contrast velocity distributions in a subsurface model. The synthetic case studies presented justify the validity and efficiency of this algorithm. The computational time is also a limitation for the use of the conventional ray-tracing methods in the tomographic applications in a larger domain. In this particular scheme the computational time being very small, we could successfully model a large domain of 512×512 grid space in a reasonable time. The resolution criterion necessitates the choice of very small cell size in the cross-hole tomographic application thereby increasing the number of cells in the region to be scanned. The present algorithm can be a right choice in this direction.

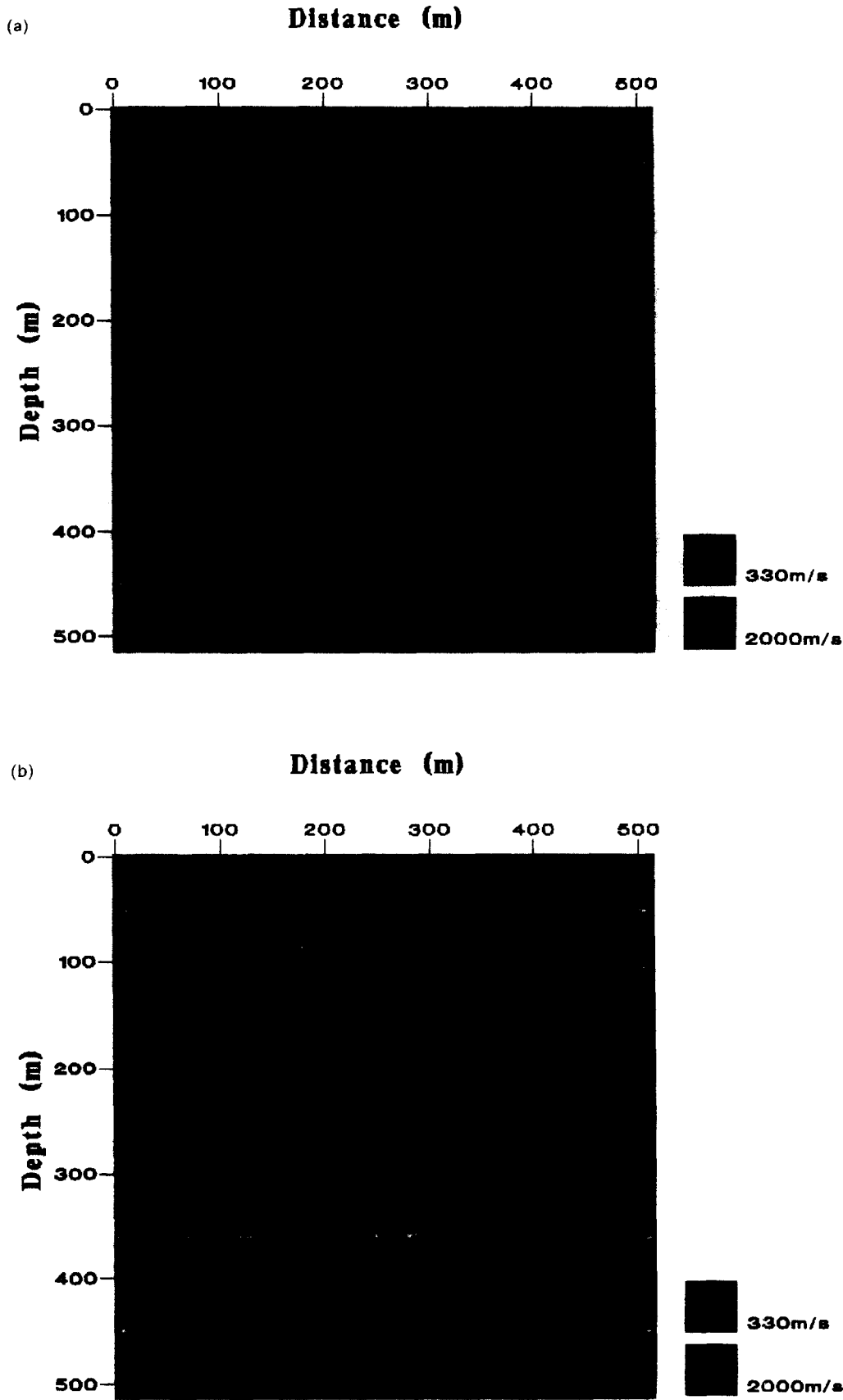


Fig. 9 (a and b)

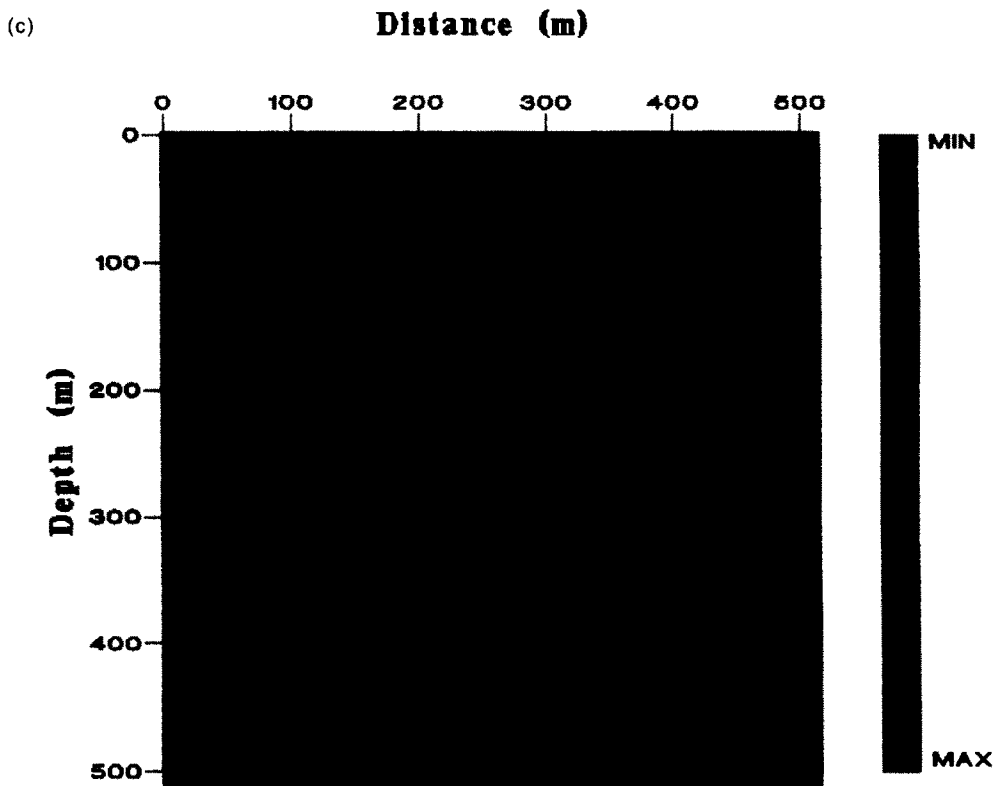


Fig. 9(c)

Fig. 9. (a) Physical configuration and superimposed ray diagram for a void in a coal seam using linear interpolation scheme. (b) Physical configuration and superimposed ray diagram for a void in a coal seam using nonlinear interpolation scheme. (c) Wave propagation snapshot of the void model of (b) for one source and four receivers.

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